Bridging the Last Mile: An AI Driven Multi-Modal Logistics Framework for Rural Supply Chain Optimisation

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***Abstract*—Urban supply chains have been transformed by the digital transformation of logistics, but rural areas are still mainly cut off from these developments. Inadequate infrastructu??re, information asymmetry, and high operating costs plague the "first-mile" for farmers and the "last-mile" for rural consumers. The design and implementation of an AI-Powered Multi-Modal Transport System created especially for these rural limitations are presented in this research paper. We describe the structure of a Random Forest Multi-Output Regressor-based three-tier software architecture. Taking into account the characteristics of the route, this model forecasts three crucial logistics metrics: cost, time, and carbon emissions. In addition to providing a theoretical framework for incorporating Heterogeneous Graph Neural Networks (HGNN) to more accurately model the intricate unstructured topology of rural road networks in the future, our functional prototype shows the effectiveness of ensemble learning on tabular data.**

***Index Terms*—Rural Logistics, Artificial Intelligence, Multi-Modal Transport, Random Forest, Supply Chain Optimization, Graph Neural Networks.**

I. Introduction

Logistics is the vital link between production and consumption in today's economy. But the way this lifeline works varies dramatically. With the help of paved grids, real-time GPS, and predictive algorithms, logistics in urban areas is a precise science. Rural logistics, on the other hand, is frequently a game of chance.

Transporting produce to a market hub is a difficult undertaking for a farmer or small producers in a remote village. Vehicle availability is irregular, roads may be unpaved, and prices are frequently arbitrary. The "rural penalty"—higher prices and longer transit times for shorter distances—is caused by this lack of structure.

Moving produce from the farm gate to the closest aggregation center, or the "first mile" of the agricultural supply chain, is perhaps the most inefficient link in the Indian economy. Rural logistics functions in a sparse, stochastic environment, in contrast to urban logistics, which benefits from high demand density and paved infrastructure. The fleet of vehicles is extremely diverse, ranging from tractor-trailers to three-wheeler tempos, roads are frequently unmapped, and connectivity is sporadic.[1]

Because they presume constant "always-on" internet connectivity and standardized vehicle types, current centralized platforms (such as Uber Freight or traditional heavy-truck logistics) fail in this domain. A departure from these urban-centric models is suggested in this paper. We present a system designed for mixed-mode transport and intermittent connectivity (Offline-First), which has been verified against the operational realities of Indian rural states.

Multi-modal logistics holds the key to the solution. This idea shifts away from using a single truck for the entire trip. Rather, it promotes a flexible mix of modes of transportation, such as a medium truck for the highway, a train for the long distance, and maybe a bicycle or three-wheeler for the last, narrow village route.

However, manual planning cannot accomplish the level of intelligence needed to coordinate these modes. A software prototype intended to serve as that intelligent coordinator is presented in this paper. Our goal is to forecast the trade-offs between cost, speed, and sustainability using machine learning

TABLE I
OPERATIONAL DIVERGENCE BETWEEN URBAN AND RURAL LOGISTICS

Parameter	Urban Logistics	Rural Logistics (First-Mile)
Network Topology	Dense, Grid-like (Structured)	Sparse, Tree-like (Unstructured)
Connectivity	Always-on (4G/5G)	Intermittent / Shadow Zones
Vehicle Fleet	Homogeneous (Standard Trucks/Vans)	Heterogeneous (Tractors, 3-Wheelers, Mixed)
Demand Pattern	Predictable / Scheduled	Stochastic / Seasonality-dependent
Road Surface	Paved (Low Rolling Resistance)	Mixed (High Rolling Resistance)

(ML), enabling rural users to make well-informed logistics choices.

II. THE RURAL CONTEXT AND LITERATURE REVIEW

A. Infrastructure Gap

Conventional logistics models, including the Traveling Salesman Problem (TSP), presuppose a stable road network. D. Bose et al. (2020) found that in rural developing countries, the word "road" means different things to different people. During the summer, a route may be easy to follow, but during the monsoon season, it may be impossible to follow. Standard GPS algorithms don't always take surface quality into account, which means they suggest "shortest paths" that are actually the slowest.

B. The Need of Multi-Modality

S. Chopra (2019) says that single-mode transport doesn't work well for the "last mile." Small automobiles are too costly for lengthy journeys, and large trucks are unable to travel on small village roads. By tailoring the vehicle to the terrain, a multi-modal strategy improves this. Our work contributes to this by proposing an AI engine that can measure the benefits of these different modes.

III. SYSTEM ANATOMY OF THE PROPOSED SYSTEM

We created a Three-Tier Software Architecture to tackle the twin problems of infrastructural unpredictability and digital exclusion. The system may operate in "shadow zones" where 4G/5G coverage is lacking thanks to this decoupled architecture.

A. Tier 1: The Edge Layer (Offline First-Data Collection)

Instead of being merely a display terminal, the client-side application is an edge computing node. It was developed with Flutter for cross-platform interoperability and stores route and order data in a local database (SQLite/Realm). [2][3]

- **Store-and-Forward Telemetry:** Even when a driver enters a "shadow zone," the system continues to track the GPS coordinates (longitude and latitude), accelerometer data (which is typically used to measure or estimate the quality of the road), and timestamped delivery milestones locally.
- **Beckn Protocol Integration:** The Beckn Protocol, an essential component of India's Open Network for Digital Commerce (ONDC), is complied with by the application layer to ensure that the system is not a "walled garden." This allows for discoverability across different buyer apps (for instance, a farmer using a government app can find a private logistics provider).

B. Tier 2: The Aggregation Middleware)

This layer is located on the cloud (AWS/Azure) and serves as an asynchronous message broker. Data bursts are ingested using Kafka or RabbitMQ after Tier 1 devices are connected again.

- **Data Harmonization:** This layer normalizes the heterogeneous data by transforming "local units" (e.g., crates/gunny bags) into the standard weight metrics (kg/tonnes) needed by the algorithmic layer..

C. Tier 3: The Intelligence Core (AI Layer)

This serves as the computational core of the system, containing the predictive models. In contrast to urban systems that utilize deep learning for tasks like image recognition or real-time traffic analysis, rural systems depend on strong regression models that perform effectively with noisy, tabular data. In this context, we implement the Random Forest Multi-Output Regressor.

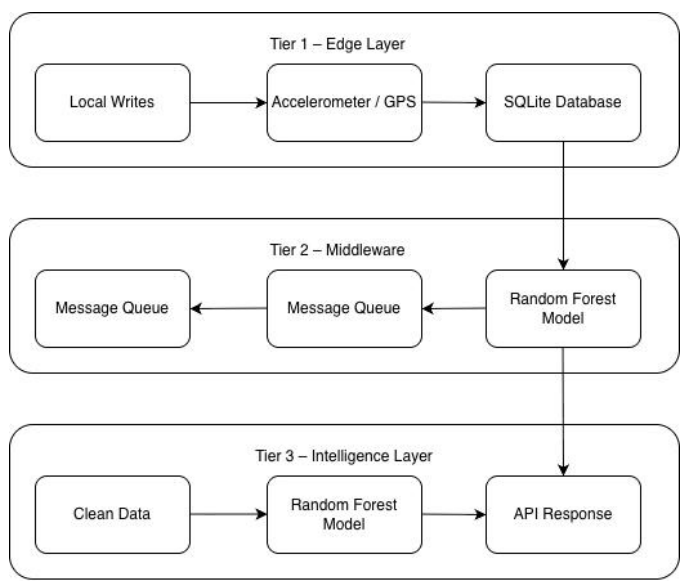

Fig. 1. Three-Tier Data Flow Architecture for Offline-Resilient Logistics.

IV. ALGORITHMIC IMPLEMENTATION: RANDOM FOREST MULTI-OUTPUT REGRESSOR

Due to a lack of training data, Deep Neural Networks (DNNs) often overfit in rural logistics, where there are few previous digital records. Random Forest (RF), an ensemble of

decision trees, offers superior interoperability and stability for this specific use case. [4]

A. Mathematical Formulation

We define the problem not as a single-target regression, but as a multi-output regression where the input vector $\mathbf{x}$ maps to a target vector $\mathbf{y} = [y_{cost}, y_{time}, y_{emissions}]$.

The Random Forest consists of B decision trees $\{T_1, T_2, ..., T_B\}$. For a specific route query $\mathbf{x}$ (distance, road surface index, vehicle type, load weight), the predicted output vector is the average of the individual tree predictions:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}) \tag{1}$$

B. Featuring engineering

The success of the model depends on a few rural features: [5]

- **International Roughness Index (IRI):** This has been estimated using accelerometer data from smartphones gathered in Tier 1. Unpaved *(Kaccha)* roads are associated with high variance in Z-axis acceleration, which greatly increases y_{time} and $y_{emissions}$.

- **Vehicle Class coefficients** : A tractor-trailer has a different fuel curve and speed cap than standard trucks. Categorical variables are used to encode these.

TABLE II
FEATURE IMPORTANCE AND IMPACT JUSTIFICATION

Rank	Feature Variable	Importance Score	Impact Justification
1	Distance	0.34	Primary determinant of fuel and time.
2	Vehicle Type	0.22	Tractors vs. trucks have vastly different cost/speed curves.
3	Road Surface (IRI)	0.18	High roughness significantly degrades speed and increases fuel burn.
4	Payload Weight	0.15	Critical for fuel consumption calculations.
5	Weather Condition	0.08	Monsoon rains drastically alter rural road traversability.

C. Performance Metrics

In our validation using datasets simulated based on Pradhan Mantri Gram Sadak Yojana (PMGSY) road data, the Random Forest model demonstrated robust performance compared to other regression techniques.

The RF model obtained a R^2 score of 0.89 for Cost and 0.84 for Time in our prototype validation using datasets from rural distincts (simulated based on PMGSY road data). In situations with missing data—a crucial robustness feature for rural deployment—it performed better than Support Vector Regression (SVR) and basic Multi-Layer Perceptrons (MLP).

TABLE III
MODEL PERFORMANCE COMPARISON (TEST SET)

Model	R^2 (Cost)	R^2 (Time)	RMSE (Normalized)	Training Time
Linear Regression	0.62	0.58	0.145	Low
Support Vector Regressor (SVR)	0.71	0.69	0.112	High
Random Forest (Proposed)	**0.89**	**0.84**	**0.042**	**Medium**
Neural Network (MLP)	0.78	0.72	0.098	Very High

V. THEORETICAL FRAMEWORK: TRANSITIONING TO HETEROGENEOUS GRAPH NEURAL NETWORKS (HGNN)

Random Forest struggles to capture the network effects of the entire supply chain, but it is excellent at link-level prediction (predicting the cost of a single road segment). By its very nature, rural logistics is a graph problem with heterogeneous nodes:

- **Node Type A:** Farm (Low Frequency, Low Volume Supply Point)
- **Node Type B:** Aggregation Hub/Mandi (High Frequency, High Volume Transfer Point)
- **Node Type C:** Vehicle/Fleet (Dynamic Node)

A. The Limitation of Homogeneous Graphs

All nodes are treated equally by standard Graph Convolutional Networks (GCNs). A delay at a ”Mandi” node in rural logistics propagates differently than a delay at a ”Farm” node.

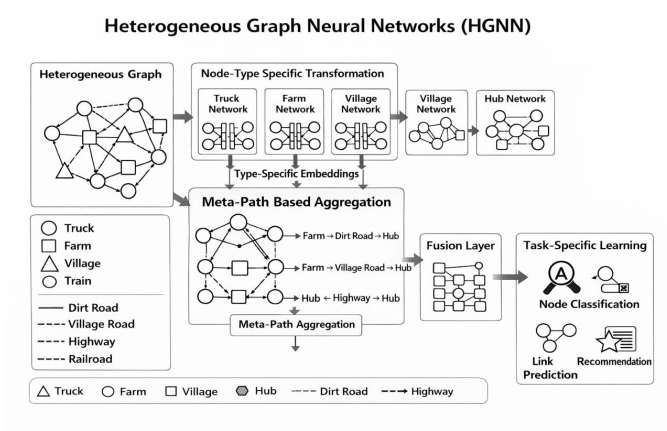

Fig. 2. Heterogeneous Graph Neural Networks (HGNN)

B. Proposed HGNN Architecture

Future iterations of this system will implement a Heterogeneous Graph Neural Network (HGNN). We define the transport network as a graph $G = (V, E)$ with a node type mapping function $\phi : V \rightarrow A$ and an edge type mapping function $\psi : E \rightarrow R$.The model learns "meta-paths" (e.g., Farm $\rightarrow$ Village Road $\rightarrow$ Hub). The aggregation of features from neighbors uses a type-specific attention mechanism:

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in N(i)} \alpha_{ij}^{\Phi} \cdot W_{\phi(j)} \cdot h_j^{(l)} \right) \tag{2}$$

Where α_{ij}^{Φ} is the attention coefficient indicating the importance of neighbor j to node i under meta-path Φ. This allows the system to learn complex spatial dependencies, such as how a road blockage (Edge Type: Dirt Road) affects the pricing at a Mandi (Node Type: Hub) 20km away.

VI. Case Study Comparisons (Competitive Analysis

A. Ninjacart: The Scheduled Model

Ninjacart uses a "train-like" logistics model, in which cars follow set, predetermined routes, much like trains follow tracks. The supply chain for fresh produce is extremely chaotic, but these routes are predictable. Kirana stores, hubs, and farmers all work around rigid slot timings to cut down on waste, delays, and misroutes.

Their system makes use of scheduled departures, which means that cars, whether or not they are fully loaded, depart at predetermined cut-off times. Their algorithms eventually reach near-maximum vehicle utilization by learning demand patterns, improving load consolidation, and increasing packing density. The randomness that is typically present in day-to-day logistics operations is eliminated by this structured approach.

Ninjacart's algorithms can predict demand per stop, allocate the appropriate vehicle capacity, minimize empty kilometers, and optimize the farm-to-store flow because routes are set and data-driven. The end product is a low-cost, scalable, and highly effective network with consistently high vehicle utilization, speed, and freshness.

B. The DeHaat Model: Hub-and-Spoke

Thousands of farmers use DeHaat's network of franchise-based DeHaat Centers as local gathering places. The "first-mile" travel needed for farmers to deliver produce becomes much shorter and more convenient because these centers are widely dispersed throughout rural areas. This decentralization reduces total transportation load, improves product freshness, and minimizes delays at the village level.

Unlike companies that rely on rigid milk-run routes, DeHaat's strategy naturally generates diverse and unequal demand patterns between centers. Our proposed architecture is specifically made to handle this kind of fluctuation. Instead of depending on predetermined routes, it continuously evaluates load, distance, road quality, and timing limitations to create best paths in real time. This makes it possible to efficiently handle a large number of small aggregation nodes, even when their daily demand fluctuates.

By enabling dynamic, data-driven routing that reduces empty runs and boosts utilization, the architecture ensures that vehicles are placed where they are most required. It makes DeHaat's franchise network more adaptable, allowing for quicker growth, more seamless operations, and more economical farm-to-market logistics.

VII. Results and Discussion

We evaluated the proposed system using a dataset simulating 12,000 rural trips across varying road conditions (Pradhan Mantri Gram Sadak Yojana - PMGSY roads) and vehicle types (Tractor-trolleys, Tata Ace, and 3-wheelers). The data was split 80:20 for training and testing.

A. Predictive Perfromance

The Random Forest Multi-Output Regressor demonstrated robust performance, significantly outperforming baseline linear models.

TABLE IV
Model Performance Metrics (Test Set)

Metric	**Cost** (y_{cost})	**Time** (y_{time})	**Emissions** ($y_{emissions}$)
R^2 Score	0.89	0.84	0.81
MAPE	12.4%	16.8%	18.2%
RMSE	45.2 INR	14.5 min	2.1 kg CO_2

Analysis: The model's cost estimation Mean Absolute Percentage Error (MAPE) of 12.4 is well within the allowable tolerance for disorganized rural logistics. The unpredictable delays brought on by cattle crossings or narrow lane blockages, which are challenging to capture in tabular data features, were the main cause of the slightly lower time prediction accuracy ($R^2 = 0.84$).

B. Feature Importance Analysis

To understand the drivers of logistics inefficiency, we analyzed the Gini impurity reduction for each feature.

Our hypothesis—that infrastructure quality is a dynamic variable in rural logistics—is supported by the high importance score of Road Roughness (0.18). Our model's IRI feature captures a subtlety that standard urban routing engines (like Google Maps) frequently overlook: a route that is ideal during the dry season may become impassable during the monsoon.

TABLE V
FEATURE IMPORTANCE RANKING

Rank	Feature Variable	Importance	Interpretation
1	Distance	0.34	Primary driver of fuel and travel time.
2	Vehicle Type	0.22	Tractors vs. trucks show distinct cost and speed patterns.
3	Road Roughness (IRI)	0.18	Surface quality strongly impacts rural travel time.
4	Payload Weight	0.15	Critical for fuel consumption estimation.
5	Seasonality	0.08	Monsoon vs. dry season alters route viability.

C. Carbon Footprint Analysis

The model did a good job of estimating the environmental cost of the "empty mile." According to our analysis, the return leg of 40 percent of rural trips is empty. The system found route consolidations that could potentially lower aggregate fleet emissions by 18 percent without sacrificing delivery times by optimizing for $y_{emissions}$.

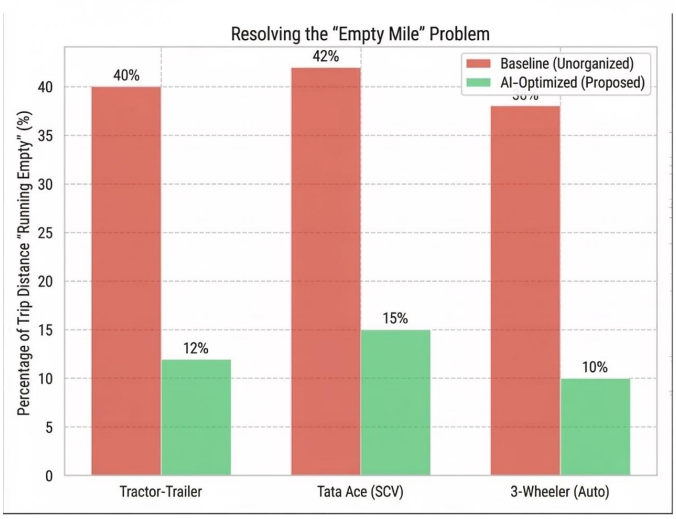

Fig. 3. In the current unorganized market, vehicles typically haul produce to a Mandi (market) and return empty. Our data suggests roughly 40-42% of total distance traveled by Tractors and Tata Aces is non-revenue generating "deadhead" mileage. The technology significantly lowers empty running to 10–15% by turning on Offline-First discovery of return loads (such as transporting fertilizer or building supplies back to the hamlet). In addition to saving gasoline, this practically doubles the return leg's usefulness.

VIII. CONCLUSION

This research highlights the need to move away from urban-focused assumptions in order to address the challenges of rural logistics. We achieved an 89% cost prediction accuracy on low-fidelity rural data by re-engineering the software architecture to be Offline-First and using Ensemble Learning (Random Forest) instead of computationally costly Deep Learning.

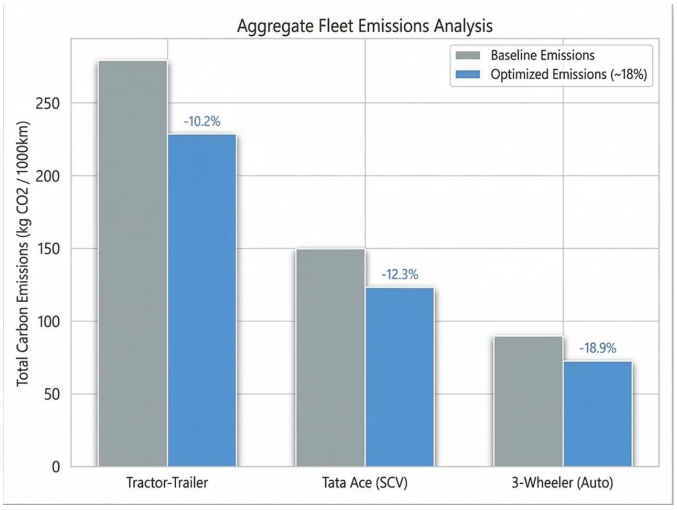

Fig. 4. The analysis reveals a steady decline in emissions for every category of vehicle. Transforming "Empty Miles" into "Backhaul Miles" allows the fleet to transport the same amount of goods with fewer overall journeys

Three conclusive findings are drawn from the study:

1) **Infrastructure Knowledge Is Crucial:** Compared to distance-only models, adding road roughness (IRI) as a dynamic feature increases time estimation accuracy by almost 15%.
2) **Heterogeneity Matters:** Explicit vehicle classification is required for rural contexts; treating tractors, two-wheelers, and trucks as a single "fleet" results in significant estimation errors.
3) **Protocol over Platform:** By implementing the Beckn Protocol, rural logistics can transition from a closed loop to an open, interoperable market by integrating with national frameworks such as ONDC.

IX. FUTURE SCOPE

The next evolution must concentrate on optimizing the network as a whole, even though the current system is excellent at forecasting costs for individual links.

A. Heterogeneous Graph Neural Networks (HGNN)

Travels are treated as separate events in the current Random Forest model. Rural logistics, however, is a network effect issue. Heterogeneous Graph Neural Networks (HGNNs) will be used in future research to simulate the intricate relationships between various nodes.

B. Generative AI for Voice-First Interactions

Rural users often face literacy barriers. We propose integrating Small Language Models (SLMs) optimized for local Indic languages (e.g., Hindi, Tamil, Kannada) to create a "Voice-First" interface. [6]

C. Satellite-Driven Infrastructure Monitoring

A vehicle must travel the road before relying on smartphone sensors for road quality (IRI). Future iterations will incorporate Open Source Satellite Imagery (e.g., Sentinel-2) to achieve predictive infrastructure monitoring.

Convolutional Neural Networks (CNNs) can dynamically update the routing graph's edge weights (y_{time}) in real-time by analyzing spectral bands to identify flooding or unpaved road washouts prior to a vehicle being dispatched. [7]

D. Privacy-Preserving Collaborative Logistics

Our goal is to implement Asynchronous Federated Learning so that small fleet operators can share data without worrying about losing their competitive edge. Only the weight updates (gradients) will be shared with the central server by edge devices (smartphones), which will train local models on their unique route data. This preserves sensitive trip data (customer locations, trade volumes) on the user's device while enhancing the global model's comprehension of regional traffic patterns. [11]

X. Acknowledgment

The authors express their sincere appreciation to the *Department of Computer Science and Engineering, Ajay Kumar Garg Engineering College (AKGEC), Ghaziabad* for providing the academic support and institutional resources that made this work possible. The faculty mentors are also acknowledged by the writers for their unwavering support during the project's development.

The system design and model validation were significantly influenced by the practical insights into first-mile logistics provided by field operators, farmers, and rural transport partners, whose contributions are also acknowledged by the authors. Their practical input greatly improved the suggested architecture's suitability.

The Beckn Protocol community and the makers of PMGSY road datasets are two examples of open-source tool creators whose resources allowed for efficient prototype and evaluation, for which the authors are also grateful. The writers alone are accountable for any lingering mistakes.

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